

PREETKUMAR DIPENKUMAR KOTAK

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[GitHub](#) | [LinkedIn](#) | [Codeforces](#)

SUMMARY

CS undergrad at SVNIT who prefers understanding a system at its lowest level before building on top of it. Drawn to problems where correctness under concurrency and edge cases actually matters, and inclined to own a build end-to-end architecture, debugging, and deployment rather than stop at a surface-level fix.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT)

CGPA: 7.95

B. Tech, Computer Science and Engineering | Expected Graduation: 2028

ACHIEVEMENTS

BIOTHON 2026 – Finalist

[GitHub](#)

Marwadi University, Rajkot | Total Teams: ~400 teams | July 2026

Built KisaanLink, a WhatsApp-based marketplace connecting Gujarati farmers to equipment rentals and daily labor, cutting out the 10-30% markup charged by middlemen on tractor rentals

SKILLS

Languages: C/C++, JavaScript, 8086 Assembly, HTML, CSS

Frameworks/Libraries: React.js, Node.js, Express.js, Tailwind CSS, discord.py

Databases & Tools: MongoDB, PostgreSQL, Supabase, Redis, Git/GitHub

PROJECTS

Calculator in 8086 Assembly

[GitHub](#)

8086 Assembly | June - July 2026 | Personal project to learn low-level/systems programming

- Built a fully functional multi-digit calculator in 16-bit 8086 Assembly supporting add, subtract, multiply, divide, modulo, power, and square root, plus HEX/binary base conversion with fractional support
- Achieved 32-bit arithmetic on the 16-bit 8086 by chaining register pairs into high/low words, and implemented fixed-point decimal math (values scaled by 100) to support decimals without an FPU
- Implemented multiplication, division, and square root from scratch using custom 64-iteration shift-and-subtract routines and a digit-by-digit base-4 integer square root algorithm
- Built a custom 80x25 text-mode UI with 2D grid-based keyboard navigation, memory store/recall/clear operations, and overflow/divide-by-zero error handling

BidKar

[GitHub](#)

Node.js, Express, MongoDB, Redis, Socket.io | May 2026 - Present | 3-person team; owned the backend | Real-time auction platform with automated settlements and mocked Aadhaar-based KYC

- Built a clustered Node.js/Express auction platform (MongoDB/Mongoose, Redis) with real-time bidding via Socket.io and a custom Redis distributed lock (NX/PX + Lua-script safe release) to prevent race conditions on concurrent bids
- Automated the auction lifecycle with node-cron jobs that auto-close expired auctions, generate settlement records (10% security deposit / 90% offline balance split), and broadcast live 'auction ended' events
- Implemented JWT + Google OAuth authentication, a mocked Aadhaar-based KYC verification flow (simulating the paid government API), Razorpay payments/wallet system, and server-side PDF invoice generation using Puppeteer and EJS templates

Clan Capital Tournament Discord Bot

[GitHub](#)

Python, discord.py, PostgreSQL, asyncpg | May - July 2026 | 5,000+ line Discord bot built for personal use, cutting manual tournament-hosting time by ~2 hours/day

- Built a Discord bot with 43 slash commands across 7 modular cogs to manage Clan Capital tournaments, supporting a Clan Capital Tournament in Clash of Clans
- Designed a 9-table relational PostgreSQL schema and implemented role-based commands (e.g. view-bases, send-bases) with custom permission checks across cog files
- Used PIL to dynamically generate match-result images
- Built a Discord anti-scam honeypot system with themed trap channels that auto-ban or kick users who post in them